

A GIS-Aware Deployment-Optimized Wireless Sensor Network for Early Forest Fire Detection

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Abstract—Wildfires are increasing in frequency and severity, while existing wireless sensor network (WSN) monitoring systems still lack deployment-aware strategies for cost-efficient coverage across heterogeneous forest environments. This paper presents a GIS-aware deployment-optimized WSN framework for early forest fire detection. The proposed approach combines low-power multi-parameter sensing with terrain-constrained hexagonal deployment, adaptive pruning, and spatial repulsion optimization to improve coverage efficiency while reducing redundancy. Simulation over an 86.25 km² forest region achieved 95.11% coverage with a terrain coverage deficit (TCD) of 2.48% and a redundancy index (RI) of 0.309, while reducing node requirements by 23.7% compared to square-grid deployment. An experimental hardware prototype further demonstrated reliable event detection with an average response latency of 0.82 s under controlled conditions. The results demonstrate the feasibility of integrating GIS-aware deployment optimization with embedded environmental sensing for scalable wildfire monitoring.

Index Terms—Wireless sensor networks, wildfire detection, coverage optimization, GIS-aware deployment, environmental sensing, cyber-physical systems.

I. INTRODUCTION

Wildfires are becoming a major environmental challenge due to climate change and rising temperatures [1]. However, conventional monitoring approaches such as satellite imaging and watchtower surveillance often suffer from delayed response, limited coverage, and high operational costs in remote forest regions [2], [3].

Wireless sensor networks (WSNs) provide a promising solution for real-time wildfire monitoring through distributed environmental sensing. Although previous studies demonstrated reliable fire detection using temperature, smoke, humidity, and gas sensors [6], [9], [10], most existing systems overlook deployment optimization and rarely consider sensing redundancy, terrain-aware adaptation, or geographic constraints [7], [8].

To address these limitations, this paper presents a GIS-aware deployment-optimized WSN framework for early forest fire detection. The proposed approach combines embedded environmental sensing with terrain-constrained hexagonal deployment, adaptive pruning, and spatial repulsion optimization to improve coverage efficiency while reducing sensing redundancy. Simulation over an 86.25 km² forest region achieved 95.11% coverage with a TCD of 2.48% and RI of 0.309, while reducing node requirements by 23.7% compared to square-grid deployment. In addition, an experimental hardware prototype achieved reliable event detection with 0.82 s response latency under controlled conditions.

This work presents a GIS-aware deployment optimization framework for wildfire monitoring integrating adaptive deployment refinement with geographic sensing constraints. The proposed framework combines hexagonal initialization, pruning optimization, and spatial redistribution to improve coverage efficiency while reducing sensing redundancy. In addition, a quantitative evaluation methodology based on spatial coverage, terrain coverage deficit (TCD), and redundancy index (RI) analysis is introduced alongside an embedded sensing prototype for environmental monitoring and mobile alert visualization.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed system architecture and detection methodology. Section IV describes the deployment optimization framework. Section V presents experimental results, evaluation, and limitations. Finally, Section VI concludes the paper and outlines future work.

II. LITERATURE REVIEW

A. Existing WSN-Based Forest Fire Detection Systems

Several wireless sensor network (WSN)-based approaches have been proposed for early forest fire detection. Alkhatib [9]

TABLE I
COMPARATIVE SUMMARY OF EXISTING WSN-BASED FOREST FIRE
DETECTION SYSTEMS

Study	Approach	Sensing	Strengths	Limitations
[6]	Energy-efficient WSN	Single temp.	Sub-network scheduling, propagation modeling, lower energy use, and extended lifetime	Temperature-only sensing and no terrain-aware deployment optimization
[9]	Fire propagation WSN	Single temp.	Multi-hop routing, Fixed spatiotemporal analysis, and simple energy-dependency efficient design	Fixed thresholds and single-sensor dependency
[10]	Hierarchical WSN with param. FWI	Multi-param.	Risk modeling, solar harvesting, high reliability, and situational awareness	High deployment cost and complexity
[11]	ZigBee-based IoT WSN	Multi-param.	Tree topology, GPRS communication, and near real-time monitoring	Communication dependency and fixed thresholds
[12]	Clustered WSN	IoT Multi-param.	Cloud integration, node-level monitoring, and scalable architecture	Cloud dependency and latency concerns

introduced a temperature-based WSN using random deployment and multi-hop communication to infer fire propagation patterns. Although computationally simple, the system relies on single-parameter sensing and fixed thresholds, limiting detection robustness. Molina-Pico *et al.* [10] proposed a hierarchical WSN architecture integrating temperature, humidity, wind, and gas sensing with Fire Weather Index (FWI)-based risk assessment. While achieving high reliability, the approach increases deployment complexity and cost.

Similarly, Alkhatib *et al.* [6] improved energy efficiency using sub-network coverage scheduling to extend network lifetime. Castro-Correa *et al.* [11] developed a ZigBee-based monitoring framework with near real-time communication, whereas Al-Dahoud *et al.* [12] proposed an IoT-enabled clustered WSN architecture for scalable monitoring. Despite their effectiveness, most existing systems rely on fixed-threshold sensing, communication-intensive architectures, or infrastructure-dependent monitoring mechanisms.

B. Research Gap Analysis

Although existing studies demonstrate reliable wildfire sensing, several limitations remain. Most existing systems focus primarily on sensing and communication reliability while overlooking deployment optimization. Sensor placement is commonly random or grid-based without considering irregular forest geometries, sensing redundancy, or terrain-aware constraints. In addition, many approaches rely on centralized communication or cloud-based monitoring, increasing complexity and limiting scalability in remote environments. Furthermore, coverage efficiency is rarely evaluated using deployment-aware metrics, leaving the trade-offs between coverage quality, redundancy, and node cost insufficiently explored. Table I summarizes the main characteristics, strengths, and limitations of existing WSN-based wildfire detection systems.

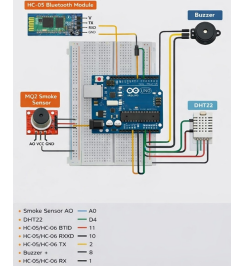


Fig. 1. Overall architecture of the proposed WSN system.

C. Motivation of the Proposed Work

Motivated by these limitations, this work proposes a lightweight and scalable wildfire monitoring framework integrating multi-parameter sensing with GIS-aware deployment optimization. Unlike conventional approaches, the proposed framework combines terrain-constrained hexagonal placement, adaptive pruning, and spatial redundancy reduction to improve coverage efficiency while minimizing node requirements. This enables more cost-efficient and geographically adaptive WSN deployment for large-scale forest monitoring.

III. PROPOSED SYSTEM DESIGN

A. System Architecture

The proposed WSN architecture consists of sensing, communication, and monitoring layers, as illustrated in Fig. 1. The sensing layer integrates MQ-2 smoke and DHT22 temperature-humidity sensors with local alert components. The prototype uses HC-05 Bluetooth for short-range communication during validation, while scalable deployment can adopt LoRaWAN or ZigBee for large-scale wildfire monitoring.

B. Hardware Components Description

The prototype node integrates sensing, processing, communication, and alert components for wildfire detection.

1) *MQ-2 Smoke Sensor*: Low-cost gas sensor for smoke and flammable gas detection requiring a 2-minute warm-up period.

2) *DHT22 Temperature-Humidity Sensor*: Provides digital temperature and humidity measurements for multi-parameter sensing.

3) *Processing Unit (Arduino Microcontroller)*: ATmega328P microcontroller handling sensing, threshold evaluation, and alert control.

4) *Wireless Communication Module*: HC-05 Bluetooth module used for short-range communication during prototype validation.

5) *Alert Actuation Components*: Passive buzzer and 5 V relay for local alert generation.

C. Fire Detection Logic

Fire detection is performed locally using a lightweight multi-parameter threshold-based model. The node continuously monitors smoke concentration $S(t)$, temperature $T(t)$, and humidity $H(t)$ to infer potential fire events.

TABLE II
DETECTION THRESHOLD PARAMETERS

Parameter	Value	Unit
Smoke threshold margin (ΔS)	40	ADC counts
Temperature threshold (T_{th})	50	$^{\circ}\text{C}$
Humidity threshold (H_{th})	30	% RH

1) *Detection Decision Formulation*: A fire event is detected when smoke concentration exceeds a baseline threshold or when elevated temperature occurs simultaneously with low humidity.

$$S_{\text{detected}} = \begin{cases} 1, & \text{if } S(t) > S_{\text{baseline}} + \Delta S \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$T_{\text{detected}} = \begin{cases} 1, & \text{if } T(t) > T_{th} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$H_{\text{low}} = \begin{cases} 1, & \text{if } H(t) < H_{th} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

The final detection decision is expressed as:

$$F = S_{\text{detected}} \vee (T_{\text{detected}} \wedge H_{\text{low}}) \quad (4)$$

where S_{baseline} is the baseline smoke level, while ΔS , T_{th} , and H_{th} denote the calibrated thresholds.

2) *Threshold Calibration*: The detection thresholds were empirically calibrated under controlled conditions. Table II summarizes the calibrated threshold parameters used during prototype validation.

3) *Detection Algorithm*: Algorithm 1 summarizes the fire detection process. The node continuously evaluates smoke, temperature, and humidity measurements to detect potential fire events and trigger local alerts when abnormal conditions are identified.

4) *Design Justification*: The proposed detection logic is designed for resource-constrained WSN environments. Lightweight threshold evaluation ensures low computational overhead, while multi-parameter sensing improves robustness against environmental fluctuations and false alarms. In addition, edge-level processing reduces dependence on centralized infrastructure and supports future energy-efficient deployment strategies.

IV. METHODOLOGY

A. Problem Formulation

Let the deployment space be represented as a discretized terrain domain $\Omega \in \mathbb{R}^{M \times N}$, where each sensor node $s_i \in S$ is positioned at coordinates (x_i, y_i) with sensing radius r_i . The objective is to maximize terrain-aware coverage while minimizing sensing redundancy.

The optimization problem is formulated as:

$$\max_S (\mathcal{C}(S) - \beta \mathcal{O}(S) - \gamma \mathcal{D}(S)) \quad (5)$$

Algorithm 1: Fire Detection and Alert Logic

Input : $S(t)$, $T(t)$, $H(t)$ sensor readings at time t
Output: Alert activation and monitoring notification

- 1 Initialize sensors and communication modules;
- 2 Perform MQ-2 warm-up for 120 s;
- 3 Compute $S_{\text{baseline}} \leftarrow$ average of 40 samples;
- 4 **while** node is active **do**
- 5 Acquire sensor readings $S(t)$, $T(t)$, and $H(t)$;
- 6 $S_{\text{detected}} \leftarrow (S(t) > S_{\text{baseline}} + \Delta S)$;
- 7 $T_{\text{detected}} \leftarrow (T(t) > T_{th})$;
- 8 $H_{\text{low}} \leftarrow (H(t) < H_{th})$;
- 9 **if** $S_{\text{detected}} \vee (T_{\text{detected}} \wedge H_{\text{low}})$ **then**
- 10 Activate buzzer and relay;
- 11 Transmit alert notification;
- 12 **else**
- 13 Deactivate alert state;
- 14 Wait 500 ms;

where $\mathcal{C}(S)$ denotes spatial coverage, $\mathcal{O}(S)$ represents sensing overlap, and $\mathcal{D}(S)$ denotes deployment imbalance.

B. Terrain Difficulty Index Construction

A Terrain Difficulty Index (TDI) was constructed by combining normalized terrain slope, ruggedness, vegetation density, and land obstruction priors:

$$TDI(x, y) = w_1 S(x, y) + w_2 R(x, y) + w_3 V(x, y) + w_4 O(x, y) \quad (6)$$

where the weighting factors satisfy:

$$w_1 + w_2 + w_3 + w_4 = 1 \quad (7)$$

C. Adaptive Radius Scaling

Sensor coverage radii were dynamically adjusted according to local terrain complexity:

$$r_i = r_{\min} + \frac{r_{\max} - r_{\min}}{1 + \exp(\alpha(TDI_i - \tau))} \quad (8)$$

where $r_{\min}$ and $r_{\max}$ denote the minimum and maximum sensing radii.

D. Overlap-Aware Optimization

To suppress sensing redundancy, pairwise overlap penalties were introduced:

$$\mathcal{O}(S) = \sum_{i \neq j} \exp\left(-\frac{d_{ij}}{r_i + r_j}\right) \quad (9)$$

where d_{ij} denotes the Euclidean distance between nodes.

Algorithm 2: AR-MSGO-TAGI Terrain-Aware Sensor Deployment

Input : Terrain layers: DEM, NDVI, WorldCover; sensor budget N

Output: Optimized deployment layout S^*

- 1 Extract terrain slope and ruggedness from DEM;
 - 2 Compute NDVI vegetation density;
 - 3 Generate land obstruction priors;
 - 4 Normalize all terrain layers;
 - 5 Construct normalized TDI map;
 - 6 Initialize uniform hexagonal deployment S ;
 - 7 **for** each sensor node $s_i \in S$ **do**
 - 8 Compute adaptive sensing radius r_i ;
 - 9 **for** each optimization iteration **do**
 - 10 Compute overlap penalty;
 - 11 Compute terrain-aware repulsive interaction priors;
 - 12 Detect uncovered high-TDI regions;
 - 13 Reinsert nodes into high-TDI uncovered regions using gap-depth prioritization;
 - 14 Update deployment topology and adaptive sensing radii;
 - 15 Perform pruning-ratio sweep;
 - 16 Select Pareto-optimal configuration;
 - 17 Evaluate Coverage, TCD, and RI;
 - 18 **return** Optimized deployment S^* ;
-

E. Terrain-Aware Global Interaction (TAGI)

Terrain-aware repulsive forces were incorporated to improve spatial redistribution:

$$P_i = \sum_{j \neq i} \frac{(TDI_j - TDI_i)}{d_{ij}^\gamma} \hat{d}_{ij} \quad (10)$$

where $\hat{d}_{ij}$ is the normalized directional vector.

F. Coverage Objective

Spatial coverage was computed as:

$$\mathcal{C}(S) = \frac{\sum_{(x,y) \in \Omega} \mathbb{I}(x,y)}{|\Omega|} \times 100 \quad (11)$$

where $\mathbb{I}(x,y)$ indicates whether location (x,y) is covered by at least one sensor node.

G. Optimization Workflow

The complete optimization pipeline of the proposed AR-MSGO-TAGI framework is summarized in Algorithm 2. Terrain-aware optimization and adaptive node redistribution are guided using a Terrain Difficulty Index (TDI) derived from DEM, NDVI, and WorldCover data.

TABLE III
EXPERIMENTAL CONFIGURATION AND SYSTEM PARAMETERS

Parameter	Value
Study Region	86.25 km ² forest ROI
Grid Dimensions	400 × 400
Sensor Budget	3529 Nodes
Baseline Deployment	Hexagonal Grid
Sensing Radius	95 m
Pruning Ratio	7.0%
Overlap Penalty (β)	1.8
Repulsion Decay (γ)	2.5

TABLE IV
TOP HYPERPARAMETER CONFIGURATIONS FROM GRID SEARCH

β	γ	Coverage (%)	TCD (%)	RI
1.8	2.5	95.11	2.48	0.309
2.0	2.5	95.09	2.50	0.309
1.8	2.0	95.03	2.56	0.310
2.4	2.5	95.02	2.55	0.310

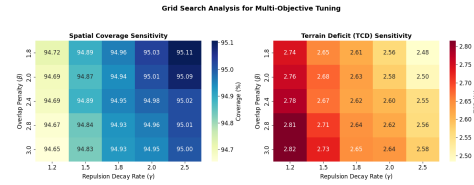


Fig. 2. Sensitivity analysis of overlap penalty (β) and repulsion decay (γ).

V. RESULTS AND DISCUSSION

A. Experimental Configuration

Experiments were conducted over an 86.25 km² forest region using terrain information derived from SRTM DEM, Sentinel-2 NDVI, and ESA WorldCover data to construct the Terrain Difficulty Index (TDI). A fixed budget of 3529 sensor nodes was used for all deployment configurations. The proposed framework applied adaptive pruning and terrain-aware redistribution, achieving the optimal configuration at a 7.0% pruning ratio with $\beta = 1.8$ and $\gamma = 2.5$. Table III summarizes the experimental parameters.

B. Hyperparameter Sensitivity Analysis

A grid search was performed over the overlap penalty (β) and repulsion decay (γ) to evaluate parameter sensitivity. The optimal configuration was obtained at $\beta = 1.8$ and $\gamma = 2.5$, achieving 95.11% coverage with 2.48% TCD. As shown in Fig. 2, increasing γ generally improved coverage and reduced terrain deficit, while neighboring configurations produced similar results, indicating stable parameter behavior.

Table IV summarizes the top-performing hyperparameter configurations.

Fig. 2 illustrates the evaluated hyperparameter sensitivity trends.

C. Pareto Frontier and Pruning Sweep

A pruning sweep from 2% to 10% was conducted to analyze the trade-off between spatial coverage, terrain coverage deficit (TCD), and redundancy index (RI). As shown in

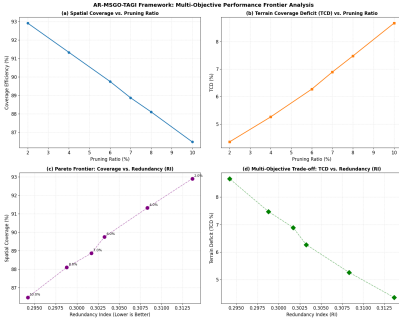


Fig. 3. Pareto frontier and pruning trade-off analysis.

TABLE V
COMPARATIVE PERFORMANCE AND ABLATION SUMMARY

Deployment Strategy	Coverage	TCD	RI
Baseline Hexagonal	93.09%	4.53%	0.181
Overlap-Aware Model	94.32%	2.75%	0.319
Repulsion-Guided Model	94.80%	2.67%	0.313
Proposed AR-MSGO-TAGI	95.11%	2.48%	0.309

Fig. 3, increasing the pruning ratio reduced coverage efficiency while lowering sensing redundancy, highlighting the balance between coverage quality and deployment compactness. The optimal operating point was obtained at a 7.0% pruning ratio with $\beta = 1.8$ and $\gamma = 2.5$, achieving 95.11% coverage, 2.48% TCD, and 0.309 RI.

D. Comparative Performance and Ablation Analysis

An ablation study was conducted to evaluate the contribution of each framework component under an identical sensor budget. Table V summarizes the comparative performance results for the baseline deployment, overlap-aware optimization, terrain-aware repulsion, and the complete AR-MSGO-TAGI framework. Progressive integration of the proposed optimization components consistently improved spatial coverage while reducing terrain coverage deficit (TCD). The complete AR-MSGO-TAGI framework achieved the best performance with 95.11% coverage and 2.48% TCD, demonstrating the effectiveness of combining overlap suppression with terrain-aware spatial redistribution.

E. Spatial Topology and Coverage Analysis

Fig. 4 illustrates the optimized deployment topology and spatial coverage analysis of the proposed framework. Fig. 4(a) shows the final terrain-aware sensor deployment, where node density increases in high-TDI regions to improve sensing reliability, while wider spacing is maintained in less constrained terrain to reduce redundancy. Fig. 4(b) demonstrates strong spatial coverage continuity across the monitored region with only limited uncovered areas. The corresponding Terrain Difficulty Index (TDI) distribution is shown in Fig. 4(c), highlighting environmentally constrained regions requiring denser sensing allocation. Finally, Fig. 4(d) presents the residual coverage gap depth map, indicating that remaining uncovered

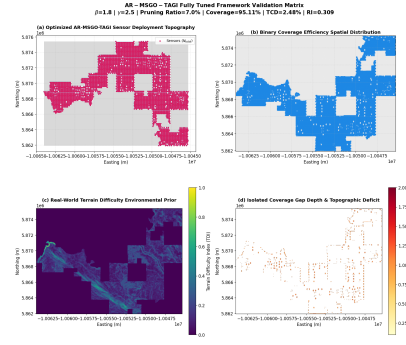


Fig. 4. Spatial deployment and coverage analysis: (a) optimized node topology, (b) binary coverage map, (c) Terrain Difficulty Index (TDI), and (d) coverage gap depth map.

TABLE VI
RESIDUAL SPATIAL GAP STATISTICS

Metric	Value (Meters)
Mean Spatial Gap Depth	47.82 m
95th Percentile Gap Distance	65.98 m
Maximum Gap Radius	93.31 m

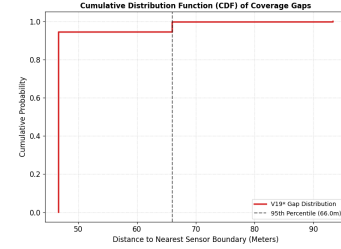


Fig. 5. CDF of residual coverage gap distances.

regions are sparse and spatially isolated. Overall, the results confirm the effectiveness of the proposed terrain-aware optimization strategy in balancing coverage efficiency, redundancy reduction, and adaptive spatial deployment.

Residual coverage gaps were analyzed using the cumulative distribution function (CDF) of uncovered pixel distances. Table VI summarizes the residual gap statistics.

Fig. 5 illustrates the corresponding CDF distribution of residual coverage gaps.

F. Statistical Robustness

A Monte Carlo analysis with 10 independent runs showed stable optimization behavior with a mean coverage of 95.09% and low standard deviation ($\sigma = 0.035$). Statistical testing also confirmed significant improvement over the baseline deployment ($p < 0.05$).

Table VII summarizes the robustness analysis and statistical metrics.

G. Computational Complexity

Table VIII summarizes the computational runtime comparison. The baseline hexagonal deployment required 0.52 s, while the proposed AR-MSGO-TAGI framework required 10.50 s due to terrain-aware optimization.

TABLE VII
ROBUSTNESS ANALYSIS AND STATISTICAL METRICS

Metric	Value
Mean Stochastic Coverage (%)	95.09
Coverage Standard Deviation (σ)	0.035
Statistical Significance	$p < 0.05$
Number of Monte Carlo Runs	10

TABLE VIII
COMPUTATIONAL RUNTIME PERFORMANCE COMPARISON

Deployment Method	Runtime (Seconds)
Baseline Hexagonal	0.52
Proposed AR-MSGO-TAGI	10.50

TABLE IX
HARDWARE PROTOTYPE CONFIGURATION PARAMETERS

Parameter	Value
Core Microcontroller	Arduino Uno (ATmega328P)
Gas/Smoke Sensor	MQ-2 Sensor
Temperature-Humidity Sensor	DHT22 Sensor
Sensor Sampling Interval	500 ms
Baseline Initialization Buffer	40 Samples
Wireless Communication	HC-05 Bluetooth Module

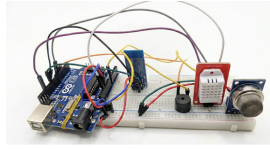


Fig. 6. Hardware prototype implementation of the proposed sensing node.

H. Prototype Hardware and System Verification

A hardware prototype of the proposed wildfire sensing node was implemented using low-cost embedded components. Table IX summarizes the hardware and software configuration parameters.

Fig. 6 illustrates the implemented hardware prototype integrating the Arduino Uno, MQ-2 smoke sensor, DHT22 sensor, relay module, buzzer, and HC-05 Bluetooth interface.

The prototype was evaluated under controlled conditions to verify sensing, alert generation, and communication functionality.

I. Mobile Monitoring and Alert Verification

A Flutter-based mobile application was developed for telemetry monitoring and alert visualization. Fig. 7 illustrates the monitoring interface of the proposed sensing prototype.

The application successfully received telemetry streams and smoke-triggered alerts through the wireless communication interface during testing.

J. Limitations

This study adopts several simplifying assumptions, including isotropic sensing coverage and fixed environmental conditions. In addition, communication constraints and energy consumption were not fully incorporated into the current deployment framework.

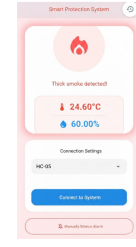


Fig. 7. Mobile application dashboard for telemetry monitoring and alert visualization.

VI. CONCLUSION AND FUTURE WORK

This paper presented a GIS-aware WSN framework for early forest fire detection integrating terrain-aware deployment optimization with embedded environmental sensing. Simulation results achieved 95.11% spatial coverage while reducing node requirements by approximately 24% compared to square-grid deployment. In addition, the hardware prototype demonstrated reliable event detection with an average response latency of 0.82 s under controlled conditions. Overall, the proposed framework demonstrates the potential of combining GIS-aware optimization with lightweight edge sensing for scalable and cost-efficient wildfire monitoring.

Future work will focus on large-scale real-world deployment validation, long-range communication integration such as Lo-RaWAN, and adaptive thresholding using lightweight machine learning techniques.

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